

Far-Western University

Faculty of Science & Technology

Mahendranagar, Kanchanpur

**Four Year Bachelor Computer Science/ General Science
(CSIT/Physical/Bio/Environment Sc./Microbio)
Entrance Examination 2074
Date 2074/08/09**

Symbol : _____ **Student's Signature :** _____

Note : Read the questions properly and write the correct option a, b, c, or d in the answer sheet provided .

Invigilator's Signature
Signature

Superintendent's

Entrance Examination 2074

F.M. : 100

Time : 2 hours

Attempt all questions

Read the following questions and write down the correct option a, b, c or d in the answer sheet provided.

Physics

20×1=20

1. What is the dimensional formula of $\frac{L}{R}$? Where L is inductance and R is resistance of the coil.
(a) [T] (b) [M] (c) [L] (d) [MLT]
2. A stone is dropped from a certain height which can reach the ground in 5 seconds. It is stopped after 3 seconds of its fall and then again released, the total time taken by the stone to reach the ground will be
(a) 6 sec (b) 6.5 sec (c) 7 sec (d) 7.5 sec
3. An airplane moving with 300 Km/hr in circular path then what is change in velocity in half revolution?
(a) zero (b) 300 Km/hr (c) 900 Km/hr (d) 600 Km/hr
4. For a given material, the Young's modulus is 2 times that of rigidity modulus, its Poisson ratio is
(a) 1 (b) 0 (c) 0.2 (d) 0.5
5. The viscosity of increases with increase in temperature.
(a) Liquid increases (b) Gas increases (c) Liquid decreases (d) Liquid and gas
6. If the earth is suddenly contract $\frac{1}{6}$ of present size without change in mass the duration of day will be nearly.....
(a) 0.67 hours (b) 5 hours (c) 6 hours (d) 4 hours
7. How much water a pump of 20 KW can raise in one minute to a height of 10m.
(a) 10000 litre (b) 12000 litre (c) 1000 litre (d) 2000 litre
8. When two electrons brought near the potential energy of the system
(a) Increases (b) Decreases (c) Remains constant (d) Zero
9. Three resistance of 2Ω are connected to form a triangle, the resistance between any two terminals is
(a) 4Ω (b) 2Ω (c) 6Ω (d) $\frac{4}{3}\Omega$
10. The ratio of specific charge of proton to that alpha particle is
(a) 1:2 (b) 2:1 (c) 1:1 (d) 4:1
11. Magnetic permeability of diamagnetic substance is
(a) Positive (b) negative (c) positive and very high (d) zero
12. x-rays are

- (a) Electromagnetic wave (b) Highly energetic (c) Absorbed by metal (d) All of above

13. Zener diode is used for....

- (a) Rectifier (b) Amplification (c) Stabilization (d) All of above

14. The appropriate percent of a radioactive substance remained after 4 half life period is

- (a) 3.5 (b) 6.25 (c) 9.25 (d) 12.2

15. All gas at same temperature has same

- (a) Kinetic energy (b) Density (c) RMS speed (d) None of above

16. As pressure increases boiling point of liquid

- (a) Increases (b) decreases (c) remains same (d) First increase then decreases

17. The efficiency of Carnot engine working between steam point and ice point is

- (a) 16.8% (b) 26.81% (c) 36.8% (d) 46.8%

18. For good audibility the reverberation time is nearly

- (a) 0.1s (b) 1s (c) 10s (d) 100s

19. Evidence of wave nature of light can not be obtained from

- (a) Diffraction (b) Interference (c) Doppler effect (d) Reflection

20. The minimum distance between an object and its real image formed by a thin convex lens of focal length f is

- (a) $2f$ (b) f (c) $4f$ (d) $f/2$

Mathematics

20×1=20

21. The perpendicular distance of the point (2, 3, -5) from the plane $x+2y-2z=9$ is

- a. $\sqrt{3}$ b. $\sqrt{3}/4$ c. $\sqrt{3}/2$ d. 3

22. Which of the following are direction cosines?

- a. 1, 0, 0 b. 2, 0, 0 c. 1, 1, 1 d. $1/2, 1/2, 1/2$

23. The conic section is said to be hyperbola if

- a. $e=1$ b. $e > 1$ c. $e < 1$ d. $e \geq 1$

24. The domain of the function $f: \mathbf{R} \rightarrow \mathbf{R}$ defined by $f(x) = \frac{1}{1-x}$ is

- a. $\mathbf{R}$ b. $(0, \infty)$ c. $\mathbf{R} - \{1\}$ d. none

25. The value of $\sin(\cos^{-1}\frac{3}{5})$ is

- a. $4/5$ b. $2/5$ c. $3/2$ d. $3/5$

26. Find the sum of the series $16+8+4+2+\dots$ is

- a. 22 b. 32 c. 36 d. 40

27. The minor of 2 in the determinant $\begin{vmatrix} 1 & 0 & 12 \\ 2 & -1 & -3 \\ 7 & 6 & 0 \end{vmatrix}$ is

- a. 72 b. 36 c. -72 d. -36

28. If w is the cube root of unity then the value of $1+w+w^2$ is

- a. 1 b. 0 c. 2 d. 3

29. The perpendiculars drawn from the vertices on their opposite sides of triangle meet at a point which is called

- a. centroid b. incenter c. circumcenter d. orthocenter

30. The equation $x^2 - 2x + 4 = 0$ have

- a. equal roots b. imaginary roots c. rational roots d. irrational roots

31. The set $\{x^2 + 4 = 0\}$ where x is real number is

- a. null set b. finite set c. singleton set d. infinite set

32. The middle term in the expansion of $(1+x)^{12}$ is

- a. 6th b. 7th c. 8th d. none

33. The identity element of any group is

- a. Unique b. not exist c. 0 d. 1

34. The sides of triangle ABC are $a=2$, $b=\sqrt{6}$ and $c=(\sqrt{3}-1)$, then

- a. 45° b. 60° c. 30° d. 90°

35. The derivative of a^x is

- a. a^x b. 0 c. $a^x \log a$ d. $a^x / \log a$

36. The value of $\int \log x dx$ is

- a. $1/x$ b. $\log x$ c. 0 d. $x \log x - x$

37. Which of the following function is discontinuous

- a. $2x+3$ b. x^2-2x+3 c. $1/x$ d. none

38. The value of $C(12,4)+C(12,5)$ is

- a. $C(13,4)$ b. $C(13,5)$ c. $C(13,6)$ d. $C(12,5)$

39. The probability of occurring the even number in the throwing a die is

- a. $1/4$ b. $1/3$ c. $1/2$ d. $1/6$

40. If r be the correlation coefficient then which of the following is true

a. $r > 1$

b. $r = 1$

c. $r = -1$

d. $-1 \leq r \leq 1$

Chemistry

20×1=20

41. The number of atoms present in 14 gm of nitrogen gas is

a) 6.02×10^{22}

b) 3.10×10^{22}

c) 1.20×10^{23}

d) 6.02×10^{23}

42. One molar solution of KMnO_4 in acidic medium is equivalent to

a) 2N

b) 3N

c) 4N

d) 5N

43. The pH of a 10^{-8} HCl solution in water is

a) 8

b) 6

c) between 7 and 8

d) between 6 and 7

44. According to Lewis concept acid is defined as .

a) An electron pair acceptor

b) An electron pair donor

c) A proton donor

d) A proton acceptor

45. Which of the following is the largest in atomic size?

a) Na

b) Mg

c) K

d) Ca

46. Structure of ammonia is

a) Pyramidal

b) Tetrahedral

c) Trigonal

d) Trigonal bipyramidal

47. The oxidation number of S in $\text{Na}_2\text{S}_4\text{O}_6$ is .

a) +10

b) +2

c) +10/2

d) +4

48. The theory of electrolytic dissociation was put forward by

a) Faraday

b) Rutherford

c) Thomson

d) Arrhenius

49. Unit of rate constant for first order reaction

a) $\text{mol}^{-1}\text{s}^{-1}$

b) s^{-1}

c) $\text{Lmol}^{-1}\text{s}^{-1}$

d) None

50. The value of Universal gas constant R is

a) $0.0821 \text{ liter atm mol}^{-1} \text{ K}^{-1}$

b) $0.082 \text{ liter atm K}$

c) $0.0821 \text{ liter}^{-1} \text{ atm K}^{-1}$

d) $0.082 \text{ liter atm}^{-1} \text{ kmol}^{-1}$

51. Malachite is an ore of

a) Zinc

b) Copper

c) Iron

d) Sodium

52. In electrolytic refining the impure metal is made

a) Cathode

b) Anode

c) Both

d) None

53. A hydrated Salt When exposed to air loses water it is called of

- a) Efflorescent b) Deliquescent c) Hydroscopic d) Hydrated

54. Tendency of catenation is highest for

- a) Carbon b) Silicon c) Tin d) Lead

55. Ozone is

- a) An allotrope of oxygen b) An isomer c) An isotope d) All of the above

56. Which of the following is an aromatic compound

- a) Benzene hexachloride b) Anthracne c) Cyclohexane d) Cyclobutadiene

57. In Lassaigne's test for nitrogen the blue color is due to formation of

- a) Ferric Ferrocyanide b) Potassium Ferrocyanide c) Sodium Ferrocyanide d) Sodium cyanide

58. The IUPAC name of $\text{CH}_3\text{OC}_3\text{H}_7$ is ...

- a) Ethoxy methane b) methoxy propane c) Methy Propyl ether d) Propylmethyl ether

59. Chloroform on warming with Ag powder gives

- a) C_6H_6 b) C_2C_4 c) C_2H_2 d) all of those

60. Isopropyl alcohol on oxidation gives

- a) Acetone b) Ether c) Ethylene d) Acetaldehyde

English

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61. It costs Rs. 100 kilo.

- a. a b. an c. the d. no article

62. Write your name the top of the page.

- a. in b. on c. over d. at

63. Be regular in the school,?

- a. can you b. won't you c. would you d. will you

64. Either you or I taking this responsibility.

- a. is b. are c. were d. am

65. The indirect speech of: David says to us, "I am very busy." Is

- a. David tells us that he is very busy.
b. David tells us that he was very busy.
c. David tells us that he has been very busy.
d. David tells us that he had been very busy.

66. Change the voice of the following: Don't throw rubbish on the road.

- a. Let the rubbish not be thrown on the ground.
- b. You are supposed not to throw rubbish on the ground.
- c. Rubbish should not be thrown on the ground.
- d. All of the given option.

67. He gave the order to leave the ship it was sinking.

- a. because of b. because c. therefore d. since

68. I don't know Rara lake is located.

- a. where b. which c. were d. when

69. He can go remain in the class.

- a. neither/nor b. either/or c. whether/or d. but/so

70. If you heat iron, it

- a. will expand b. expands c. would expand d. would have expanded

71. I will get a letter

- a. wrote b. write c. written d. to write

72. She has her food at home.

- a. cooked b. cooks c. to cook d. to be cooked

73. The synonym of 'uniqueness' is

- a. inclusive b. match c. commonness d. individuality

74. The antonym of 'emphatically' is

- a. forcefully b. easily c. indefinitely d. consistently

75. The word 'development' takes the stress on syllable.

- a. First b. second c. third d. fourth

76. She has been learning Japanese 2008.

- a. for b. in c. since d. by

77. One of the boys not done his homework today.

- a. has b. have c. had d. did

78. Had I been to Pokhara, I the Phewa Lake.

- a. would see b. will have seen c. could see d. would have seen

79. Either Varsha or you solved the paper.

- a. have b. had c. has d. were

80. The injured boy died while he

- a. was being treated b. was treated c. is treated d. were being treated

Biology(fornBio/Environmental group only)

20×1=20

81. The infective stage of the *Plasmodium* is

- a) *Anopheles* b) Merozoites c) Gametocytes d) Sporozoites

82. *Paramecium* is

- a) Ammonotelic b) Ureotelic c) Uricotelic d) Guanotelic

83. The dark colour of earthworm is due to the presence of

- a) Hemoglobin b) Haemerythrin c) Haemocyanin d) Porphyrin

84. The largest blood corpuscles are

- a) Erythrocytes b) Leucocytes c) Thrombocytes d) all of above

85) The scientific name of glass-rope sponge is

- a) *Sycon* b) *Leucosolenia* c) *Hyalonema* d) *Spongilla*

86. Which of the following species is extinct from Nepal ?

- a) Pink headed duck b) Assamese monkey c) Bengal florican d) Golden monitor lizard

87. The site of photosynthesis is

- a) Grana b) Stroma c) ER d) cytoplasm

88. Which is sun-shine vitamin ?

- a) Vit A b) Vit B c) Vit. C d) Vit . D

89. Adam's Apple is found in

- a) Head b) Neck c) Trunk d) Tail

90. Na^+ & Cl^- reabsorption takes place in

- a) PCT b) DCT c) Loop of Henle d) CD

91. The largest organ in the body is

- a) Liver b) Heart c) Intestine d) Skin

92. Montoux test is done for

- a) TB b) Typhoid c) AIDS d) Kala-azar

93. Which hormone is known as emergency hormone ?

- a) GH b) Insulin c) Thyroxin d) Adrenaline

94. is monosaccharide

- a) Fructose b) Sucrose c) Lactose d) Starch

95. Bivalents are produced in

- a) Leptotene b) zygotene c) pachytene d) Dikinesis

96. The baking yeast is

- a) *Saccharomyces* b) *Aspergillus* c) *Penicillium* d) *Agaricus*

97. The major cause of ozone depletion is

- a) CO_2 b) SO_2 c) NO_2 d) CFC

98. Cell turgidity is caused by

- a) Endosmosis b) exosmosis c) plasmolysis d) diffusion

99. The term mutation was introduced by

- a) Mendel b) Hugo de vries c) Darwin d) Lamarck

100. Which one is a sex-linked disease ?

- a) Albinism b) Beri- Beri c) Colour blindness d) Tylosis

Computer Science (for CSIT only)

20×1=20

101. Touch screen is device.
a. output b. input c. memory d. processing
102. The term 'Computer' is derived from.....
a. Latin b. German c. French d. Arabic
103. Who is the father of Computer?
a. Allen Turing b. Charles Babbage c. Simur Cray d. Augusta Adaming
104. Which device is required for the Internet connection?
a. Joystick b. Modem c. CD Drive d. NIC Card
105. ALU is
a. Arithmetic Logic Unit b. Array Logic Unit c. Application Logic Unit d. None of above
106. WAN stands for.....
a. Wap Area Network b. Wide Area Network c. Wide Array Net d. Wireless Area Network
107. MICR stands for
a. Magnetic Ink Character Reader b. Magnetic Ink Code Reader c. Magnetic Ink Cases Reader d. None
108. Junk e-mail is also called
a. spam b. spoof c. sniffer script d. spool
109. A computer cannot 'boot', if it does not have the
a. Compiler b. Loader c. Operating System d. Assembler
110. First page of Website is termed as-
a. Homepage b. Index c. JAVA script d. Bookmark
111. DOS stands for
a. Disk Operating System b. Disk operating session c. Digital Operating System d. Digital Open system
112. Who is the chief of Microsoft?
a. Babbage b. Bill Gates c. Bill Clinton d. none of these
113. 1 Byte =?
a. 8 bits b. 4 bits c. 2 bits d. 9 bits
114. An area of a computer that temporarily holds data waiting to be processed is.....
a. CPU b. Memory c. Storage d. File
115. is the key we use to run the selected command.
a. SHIFT b. TAB c. ENTER d. CTRL
116. RAM is an example of
a. Secondary memory b. Primary memory c. Main memory d. Both (b) and (c)
117. To access properties of an object, the mouse technique to use is-
a. Dragging b. dropping c. right-clicking d. shift-clicking
118. The process of transferring files from a computer on the Internet to your computer is called
a. Downloading b. uploading c. FTP d. JPEG
119. Which of the following is not a binary number?
a. 001 b. 101 c. 202 d. 110
120. first computer used in Nepal is
a. IBM1401 b. IBM 1402 c. IBM 1403 d. CDC 3600